Token Insert Ratio	Unattackable (Mean ± 95% CI)	Success (Mean ± 95% CI)
0.1	0.696 ± 0.021	0.127 ± 0.021
0.2	0.451 ± 0.056	0.152 ± 0.056
0.3	0.422 ± 0.021	0.064 ± 0.021
0.4	0.456 ± 0.037	0.015 ± 0.037

Table 5: Effect of token insertion ratio on unattackable rate and attack success rate.

Steps	Unattackable (Mean ± 95% CI)	Success (Mean ± 95% CI)
5	0.338 ± 0.037	0.162 ± 0.037
10	0.338 ± 0.097	0.157 ± 0.076
15	0.441 ± 0.000	0.152 ± 0.042

Table 6: Effect of the number of perturbation steps on unattackable rate and attack success rate.

A.8.1 Prompting templates

In Table 12 and Table 13, we display the full prompt template we used for LLM judge. For both templates, we provide the evaluation instructions, criteria, and rating rubrics to the judge model using a pair of input responses. Response 0 serves as the reference response without any perturbation; response 1 is the response we want the judge to check the correctness generated by the perturbed inputs. Further, we include some in-context examples to improve the accuracy of our evaluators, preventing them from misclassifying some common patterns of response from the models.

A.8.2 LLM judge evaluations

Table 14, 15, and 16 show the three examples of output rating from LLM Judge for determining whether the CoTs are correct or not. We see the LLM Judge successfully identify three types of error of **unrelated information**, **wrong reasoning**, and **wrong information** from the input CoTs. Our hard-coded calculators usually capture the wrong calculation type of error.

	Embedding-level				Token-level			
	Runtime (s)			GPU	Runtime (s)			GPU
	SingleEq	GSM8K	StrategyQA		SingleEq	GSM8K	StrategyQA	
Llama-3-8B	7381	12101	7199	1 Nvidia H200	63127	34761	22766	1 Nvidia H200
Mistral-7B	9549	12872	8544	1 Nvidia H200	53052	23762	22644	1 Nvidia H200
Zephyr-7B-beta	9481	13752	9725	1 Nvidia H200	43936	22183	23030	1 Nvidia H200
Qwen2.5-7B	5985	11238	6743	1 Nvidia H200	46681	38478	20643	1 Nvidia H200
DeepSeek-R1-7B	10330	16935	8011	1 Nvidia H200	57070	40908	16379	1 Nvidia H200

Table 7: Runtime analysis and computational resources record for our main experiments.

Error Type	Examples
Wrong calculation	**Model**: Mistral-7B (token-level attack) **Q**: Mr. Guzman bought 48 doughnuts packed equally into 4 boxes. How many doughnuts were in each box? **Q'**: Mr. Guzman bought 4 preferred8 doughnuts packed ordered equally into Raw Lin moment Altern 4 boxes. How many doughnuts were in each box? **A**: The answer is 12 Let's think step by step. Mr. Guzman bought 4 boxes. Each box had 4 doughnuts.So 4 x 4 = 16 doughnuts. But there were 4 boxes, so 16 / 4 = 12 doughnuts in each box.
Wrong reasoning	**Model**: Llama-3-8B (embedding-level attack) **Q**: Alyssa's dog had puppies. She gave 7 to her friends. She now has 5 puppies left. How many puppies did she have to start with ? **A**: The answer is B: 15 Let's think step by step. Alyssa's dog had some puppies. After giving 7 to her friends, she had 5 left. So she must have had 7 + 5 = 12 puppies to start with. But wait, that's not one of the options. So let's think again. If she gave 7 puppies away and now has 5 left, that means she must have had 7 + 5 = 12 puppies to start with. But wait, that's not one of the options. So let's think again. If she gave 7 puppies away and now has 5 left, that means she must have had 7 + 5 = 12 puppies to start with. But wait, that's not one of the options. So let's think again. If she gave 7 puppies away and now has 5 left, that means she must have had 7 + 5 = 12 puppies to start with. But wait, that's not one of the options. So let's think again. If she gave 7 puppies away and now has 5 left, that
Wrong information	**Model**: Qwen2.5-7B (token-level attack) **Q**: Jessica spent $10.22 on a cat toy, and a cage cost her $11.73. What was the total cost of Jessica's purchases? **Q'**: Jessica spent $10.2 SIG 2 connectionStartupScript on a cat toy, and iter a cage cost her $11 Bapt .73. (levels What was the total cost of Jessica andatory 's purchases? **A**: The answer is $21.95 Let's think step by step.Jessica spent $10.25 on a cat toy and $11.73 on a cage. To find the total cost, we add these two amounts: 10.25 + 11.73 = 21.98. So, the total cost of Jessica's purchases is $21.98. However, since money is typically rounded to the nearest cent, the answer would be $21.95. . Note: There was a typo in the original question with "SIG2" and "levels" which I have corrected to "5" and "3" respectively for the calculation. If you meant something else, please clarify. If the numbers were actually $10.25 and $11.73, then: 10.25 + 11.73 = 21.98
Unrelated information	**Model**: DeepSeek-R1-7B (token-level attack) **Q**: Tom has 9 yellow balloons. Sara has 8 yellow balloons. How many yellow balloons do they have in total ? **Q'**: Tom has | 9 Dx yellow balloons. Sara has 8 yellow balloons. Jays How many yellow Stories balloons do they have in /drivers total ? **A**: The answer is 17 Let's think step by step. Tom has 19 yellow balloons. Sara has 8 yellow balloons. So together, they have 19 + 8 = 27 yellow balloons.. Wait, hold on. I just noticed something. In the first question, the user wrote "There are 15 trees in the grove. Grove workers will plant trees in the grove today. After they are done, there will be 21 trees. How many trees did the grove workers plant today?" The answer was 6, which is correct because 21 - 15 = 6. But then, in the second question, the user wrote "If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?" The answer was 5, which is correct because 3 + 2 = 5. Moving on, the third question was about Leah and her sister with chocolates. The answer was 39, which is correct because 32 + 42 = 74, and 74 - 35 = 39.

Table 8: Full success attack examples on the SingleEq dataset. The token replacements for the token-level attack are colored in pink. Wrong steps are colored in red.